

Guide

#1.

Ans 1. To keep notebook lightweight as fetching nine tiles means you only download exact area you need, while satellite images are enormous — a high resolution .tif file of even small city can be several gigabytes.

Ans 2 ~~at~~ Each tile covers less area, so image is sharper and small rooftops are more visible. Also roughly we need 4x more tiles, so download takes longer.

Ans 3 Lowering confidence to 0.4 would increase our building count significantly because the model ~~includes~~ includes detections it is less sure about. This risks counting fake buildings — trees, shadows, cars mistakenly identified as rooftops. Risking to 0.3 would give us very few but highly reliable detections, but we would miss many real buildings.

Ans 5. When we fine tune, model already knows how to detect basic shapes, edges and patterns from its previous training. We just need to adjust its understanding to recognise Indian Rooftops specifically.

Ans 6 mAP is a very strict metric that compares our detections exactly against the 'answer key' label from Google Open Buildings. However, the open Building label for ducknow may themselves be imperfect or incomplete in some areas. The images look good because our eyes can see the boxes are sitting on real rooftops - the no is misleading because answer key, itself has gaps.

Ans 7 Building count treats every building same but buildings are very different in how many people they house. Our density score is wrong if we count 20 floor apartment and 1 story but as same.

Ans 8 I don't think, it was a waste of time, even though open Buildings gave a better result. Building and fine tuning our model taught us exactly how the detection pipeline works - from data preparation to training to evaluation - which we could have learned just ~~say~~ by just querying a database.

code

1. The sandbox model scaled well in session 1, because it was tested on the same kind of data it was trained on — large, regular, well separated buildings for US & European cities.

Such new buildings are completely different — small, densely packed, irregular shaped and touching each other.

A model can only recognise what it has been shown earlier. ~~That~~ That is why it failed,

2. Buildings detected = ~~6436~~ 4636
It misses mostly in densely packed areas, where rooftops are too close and also small buildings, partially occupied rooftops, etc.

3 Google Open Buildings found more buildings because it is a large, well trained and validated dataset. My model misses many small and densely packed buildings. A data scientist would not use an existing dataset when it is accurate and reliable.